

Why CUA Failures Are Hard to Diagnose

REAL CASE

Task (abridged): Find unpaid tuition e-mails and add them to the receiver field

Environment: LibreOffice Calc → Thunderbird

A Failure trajectory

1 after t=5 Source record

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Name	Email	Payment
Alice	alice@university.edu	Yes
Bob	bob@university.edu	Yes
Carl	carl@university.edu	Yes
Dan	dan@university.edu	Yes
Eve	eve@university.edu	Yes
Fox	fox@university.edu	No
Gina	gina@university.edu	Yes
Harry	harry@university.edu	Yes
Irene	irene@university.edu	Yes
James	james@university.edu	Yes
Kathy	kathy@university.edu	Yes
Lily	lily@university.edu	Yes
Mike	mike@university.edu	Yes
Nancy	nancy@university.edu	No
Olivia	olivia@university.edu	Yes
Peter	peter@university.edu	Yes
Quinn	quinn@university.edu	Yes
Rose	rose@university.edu	Yes
Stella	stella@university.edu	No
Tina	tina@university.edu	Yes

No: Fox · Iron · Nancy · Stella

The payment column identifies exactly four unpaid students; row 12 is Kathy with Yes.

2 t=6 Root reasoning

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Name	Email	Payment
Alice	alice@university.edu	Yes
Bob	bob@university.edu	Yes
Carl	carl@university.edu	Yes
Dan	dan@university.edu	Yes
Eve	eve@university.edu	Yes
Fox	fox@university.edu	No
Gina	gina@university.edu	Yes
Harry	harry@university.edu	Yes
Irene	irene@university.edu	Yes
James	james@university.edu	Yes
Kathy	kathy@university.edu	Yes
Lily	lily@university.edu	Yes
Mike	mike@university.edu	Yes
Nancy	nancy@university.edu	No
Olivia	olivia@university.edu	Yes
Peter	peter@university.edu	Yes
Quinn	quinn@university.edu	Yes
Rose	rose@university.edu	Yes
Stella	stella@university.edu	No
Tina	tina@university.edu	Yes
Tom	tom@university.edu	Yes

agent says Kelly; omits Fox

The agent invents **Kelly** at row 12 and drops Fox, even though the visible row is Kathy / Yes.

3 t=10 Recipient entry

type action: I / K / N / S

The four-address action carries the false belief into Thunderbird; the complete chips are visible at t=12.

4 1 more action

5 t=12 Final state

terminal reward 0

The draft still contains Kelly and still omits Fox when the trajectory declares completion.

B Why the root step is subtle

1 Row-level evidence

- 7 Fox = No
- 12 Kathy = Yes
- No Kelly row

Name, e-mail, and payment status must stay bound by row.

2 Perception to belief

- Visible source:
UI Fox / No; Kathy / Yes **record**
- Agent response:
R Kelly; no Fox **P1**

The root is the step-6 reasoning, not the scroll action executed afterward.

3 Intended vs. actual

- Human reference
F/I/N/S; Fox / Iron / Nancy / Stella
- Agent response - P1
I/K/N/S Iron / Kelly / Nancy / Stella

C Why the final symptom is insufficient

Expected recipients

Fox · Iron · Nancy · Stella

Observed at t=12

Iron · Kelly · Nancy · Stella

The receiver field proves the list is wrong, but the final frame alone cannot show where Kelly came from.

Hypotheses compatible with the wrong receiver list (which one is correct?)



Hallucinated list at t=6
RECORDED ROOT · P1



Source spreadsheet wrong



Thunderbird entry bug

D Three evidence challenges illustrated by this case



1 Dense table reading

Names, e-mails, and payment states must be joined across rows.



2 Visual hallucination

A plausible name is invented while a visible unpaid row is omitted.



3 Cross-app propagation

The false belief later becomes an incorrect Thunderbird recipient list.